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Rolls-Royce

ENGINEERING SPECIFICATION

RRES 90004

Revision C

First Issued 26 Mar 09

Revised 17 Jan 2017

Explanation of Definition Terms and Tolerance Interpretation

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1 SCOPE:

1.1 This specification interprets terms and practices used on Rolls-Royce Definitions.

2 REASON FOR USE:

2.1 This specification applies to any Rolls-Royce Definition on which it is invoked.

3 APPLICABLE DOCUMENTS:

3.1 The following publications form a part of this document to the extent specified herein. The applicable issue shall be the current issue, unless a specific document issue is specified. When the referenced document has been cancelled and no superseding document has been specified, the last published issue of that document shall apply. Additionally, contact your sector's Rolls-Royce Technical Authority to obtain guidance/approval on action to remove or replace the cancelled reference by a drawing, standard, and/or specification change. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

3.2 American National Standards Institute Publications, available from ANSI, 25 W. 43rd St., New York, NY 10036, or www.ansi.org:

ANSI/IEEE 315 Electrical and Electronics Diagrams, Graphic Symbols for

3.3 American Society of Mechanical Engineers Publications, available from ASME, 22 Law Dr., P.O. Box 2900, Fairfield, NJ 07007-2900, or www.asme.org.

ASME Y14.5M-1994 Dimensioning and Tolerancing

ASME Y14.38 Abbreviations and Acronyms for Use on Drawings and Related Documents

ASME Y14.41 Digital Product Definition Data Practices

4 DEFINITIONS, ACRONYMS AND SYMBOLS:

4.1 Definitions:

Blend:
The term blend used in this document is a requirement specifying a smooth transition where there is a mismatch of surfaces.

Component Definition:
A component definition is a technical definition of an item against which the item is manufactured and/or procured.

Definition:
The physical or functional requirements of an item that is disclosed directly or by reference using graphic or textual presentations, or by combinations of both.

Drawing:
A type of component definition that contains item definition on 2D sheets (paper). In reference to ASME standards the term "drawing" is synonymous with Component Definition.

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Flaw:
A flaw is any unintentional, unexpected, or unwanted interruptions in the topography typical of a surface.

Inflection
An inflection is the point of change in the curvature of an arc or curve between concave and convex.

Mismatch:
A mismatch is any offset of adjacent surfaces typically caused by misalignment and/or unintended freedom of movement of the fabrication equipment or tooling.

Reversal:
A reversal is a condition where a segment of a surface forms an arc whose centre point would be on the opposite side of the centre of an arc that is tangent to the adjacent surfaces. See Figure 1.

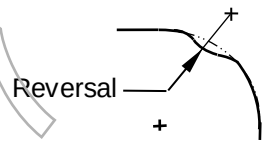

FIGURE 1 - REVERSAL

4.2 Acronyms:

All acronyms used on Rolls-Royce drawings appear in either ASME Y14.38 or this list.

Component Definition	Term
BIHEX	bi-hexagonal
CBC	compressor blade contour
CCF	Conformance Control Feature
CIP	compressor inlet pressure
CIT	compressor inlet temperature
CPS	cycles per second
EBL	Engine Buttock Line
ECM	electro-chemical machining
FPMU	fuel pump metering unit
KCF	key characteristic features
kN	kiloNewton
kPa	kiloPascal
KPC	key product characteristic
ksi	kilopound per square inch
m	metre
MGT	measured gas temperature
mm	millimetre
MPa	MegaPascal
Nm	Newton metre
PLM	Product Lifecycle Management
TPI	threads per inch

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4.3 Symbols:

4.3.1 Symbols used on component definitions shall be interpreted as follows:

12 (Number within a triangle) Flag Note Symbol: Denotes a flag note.

B (Letter within a circle): Denotes the location of a component definition revision.

5 (Number within a circle): Represents the assembly on ground support definitions, assembly tool definitions, and test equipment definitions.

2B (Number and letter within a circle): Represents individual details making up the assembly on ground support definitions, assembly tool definitions, and test equipment definitions.

3 (Number within a circle with an arrow): Denotes a Find Number. The find number is cross reference to the item number of the parts list where the item name and identification number are given.

18.5 (Decimal number within a circle or oval): Denotes an engine station line, water line, or buttock line location.

4.3.2 Electrical and Electronic Symbols used on component definitions shall be interpreted per ANSI/IEEE 315.

5 ORDER OF PRECEDENCE OF COMPONENT DEFINITION REQUIREMENTS:

The following lists the order of precedence for requirements included on component definitions. When requirements appear to be in conflict, the highest priority requirement shall take precedence.

1st (Highest) Priority	Dimensions and tolerances shown
2nd	Notes
3rd	Specifications invoked in item definition
4th	Specifications invoked in the title block (excluding RRES 90004)
5th	This specification (RRES 90004)
6th	Specifications invoked by RRES 90004
7th	Dimensions and tolerances derived from the electronic Master Model in accordance with ASME Y14.41

6 SPECIFICATION REQUIREMENTS:

6.1 Unless otherwise specified, the latest published version of each specification, definition, or model invoked on the definition is the version that applies. However, when a specification is invoked that identifies a specific revision letter or a specific revision date (e.g., ASME Y14.5M-1994), that specific version applies.

6.2 When a specification indicates that it is cancelled, obsolete, inactive, or non-current and indicates no superseding specification, the requirements of the cancelled, obsolete,

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inactive or non-current specifications still apply. When a specification indicates that it has been superseded, the requirements of the superseding specification apply.

7 COMPONENT DEFINITION RELEASE ISSUES:

P – Indicates a preliminary issue of the component definition for information purposes only.

M – Indicates a preliminary issue of the component definition for manufacturing buy-off.

F – Indicates a final issue of the component definition.

REVISION HISTORY					
ZONE	REV		DESCRIPTION	DATE	APPROVAL AUTHORITY
ALL	1	P	PRELIMINARY ISSUE UNCHECKED	21 FEB 2007	FW12345/ CDG/001
ALL	2	M	FOR MANUFACTURING ACCEPTANCE ASSESSMENT	15 MAR 2007	FW12345/ CDG/002
ALL	C	F	FINAL ISSUE ACCEPTED BY MANUFACTURING	3 APR 2007	SLN018596

FIGURE 2 – REVISION HISTORY BLOCK EXAMPLE

IMPORTANT NOTE - Parts shall not be delivered to Rolls-Royce against a **P** or **M** preliminary issue, except with written authorization from Rolls-Royce.

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8 DIMENSIONS AND TOLERANCES:

8.1 All Dimensioning and Tolerancing shall be interpreted per ASME Y14.5M-1994. The following are notable requirements, additions and exceptions:

a) Significant Datum References in Note Form – "…RELATIVE TO DATUMS A, B AND C" shall be interpreted to mean "…relative to primary datum A, secondary datum B, and tertiary datum C."

b) Unless otherwise specified, a default runout tolerance specified in a general note will not apply to features less than 2.5 millimetres in axial length, edge breaks or chamfers.

c) All specifications used on definitions, including those for material, dimensions, tolerances, surface texture and processes, apply to the finished condition of the part, unless otherwise specified. Peened or coated surfaces are an exception such that the dimensions, tolerances and surface texture apply prior to peening or application of coating(s), unless otherwise specified.

d) Dimensions or other specifications are not to be construed as dictating the use of particular manufacturing or inspection methods unless specifically expressed on the definition.

e) Tolerances and surface texture requirements established by industry or government standards shall apply to bar, sheet, tubing, and standard structural shapes when the item remains in the "as-furnished" condition on the finished part.

f) All surfaces or line elements must lie within the specified tolerance zones. All variations from the true profile (within the tolerance zones) are acceptable, unless otherwise specified.

g) Surface defects (flaws or mismatches) are not considered during inspection of surface texture.

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9 KEY PRODUCT CHARACTERISTICS (KPC):

9.1 KPC is an identified design characteristic which if non-conforming may cause 1) hazardous unsafe operating conditions for personnel using, maintaining, or depending on the product, or 2) an operational or functional failure of the product and/or an impact to the satisfactory operation of the product. Requirements for each KPC type are defined in the appropriate quality system relevant for the product or production area.

Note: In addition to the usual quality system requirements KPCs may have one or more additional requirements associated with them. For example, a critical characteristic may have restrictions on sample inspection or a KCF may have additional data capture and retention requirements.

9.2 Key Characteristic Feature (KCF):

9.2.1 KCFs are geometric product features that strongly influence the performance or service life of the product, and reduced variation from the nominal value reduces variation in the performance of the product. KCFs are measureable and can be directly influenced by process control methods at their point of manufacture or assembly.

9.2.2 Features designated as KCF in the field of a Definition and have an associated general note. See Figure 3.

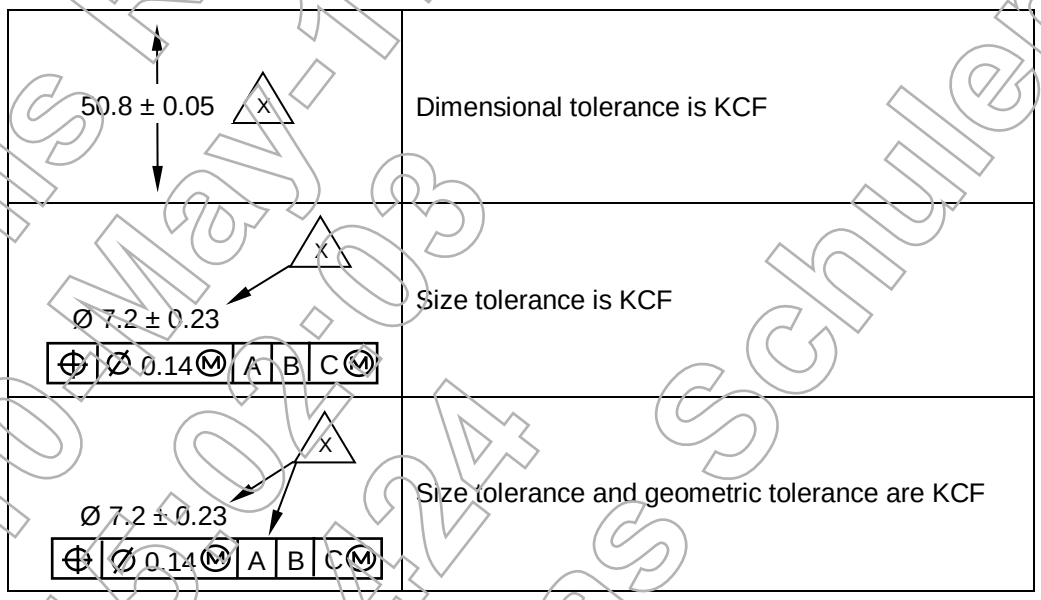

FIGURE 3 – KCF DESIGNATED FEATURES

9.3 Critical and Significant Characteristics

9.3.1 Critical characteristics are those whose primary failure is shown by the failure analysis as likely to directly result in hazardous effects. Therefore the likelihood of failure needs to be controlled to an acceptably low probability of occurrence. Significant Characteristics

9.3.2 Significant characteristics are those that are not critical but whose primary failure is shown by the failure analysis as likely to result in a substantial effect. Therefore the likelihood of failure needs to be controlled to an acceptably low probability of occurrence.

9.3.3 Features designated as Critical or Significant in the field of a Definition and will include a classification symbol. See Figure 4 and Table 1.

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50 ± 0.05 ⊕	Dimensional tolerance is critical
Ø 50 ± 0.02 ⊕ ⊕ | Ø 0.1 Ⓜ | A | B | C Ⓜ	Size tolerance is critical
Ø 50 ± 0.02 ⊕ ⊕ | Ø 0.2 Ⓜ | A | B | C Ⓜ	Size tolerance and geometric tolerance are critical
Ø 50 ± 0.02 ⊕ X ⊕ | Ø 0.2 Ⓜ | A | B | C Ⓜ	Size tolerance and geometric tolerance are critical. In addition, size and the geometric requirement is also monitored as KCF as designated by the associated note flag

FIGURE 4 – KPC DESIGNATED FEATURES

TABLE 1 – KPC SYMBOLS

SYMBOL	CLASSIFICATION	DESCRIPTION
⊕	Critical	Critical characteristics are those whose primary failure is shown by the failure analysis as likely to directly result in hazardous effects. Therefore the likelihood of failure needs to be controlled to an acceptably low probability of occurrence. Note: Critical features are not eligible for any sample plans
⊖	Significant	Significant characteristics are those that are not critical but whose primary failure is shown by the failure analysis as likely to result in a substantial effect. Therefore the likelihood of failure needs to be controlled to an acceptably low probability of occurrence.
None	Unclassified	Characteristics whose failure does not satisfy the definition of the higher classifications such as Critical or Significant Characteristics.
Flag Note	KCF	See KCF

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9.4 Conformance Control Feature (CCF):

9.4.1 CCFs are measureable features of products whose variation (within tolerance) has a significant influence on performance or service life.

9.4.2 Features are designated as CCF in the field of a Definition and have an associated general note. See Figure 5.

50.8 ± 0.05 X	Dimensional tolerance is CCF
X Ø 7.2 ± 0.23 ⊕ Ø 0.14 Ⓜ A B C Ⓜ	Size tolerance is CCF
X Ø 7.2 ± 0.23 ⊕ Ø 0.14 Ⓜ A B C Ⓜ	Size tolerance and geometric tolerance are CCF

FIGURE 5 – CCF DESIGNATED FEATURES

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10 EDGE TREATMENTS; SHARP EDGE, EDGE BREAK, RADIUS, CHAMFER, AND BLEND

10.1 All surfaces must lie within tolerance zones specified and all loose or adhering burrs shall be removed.

10.2 Sharp Edge:

10.2.1 Where a “Sharp Edge” is specified, all burrs shall be removed and the surfaces which make up a sharp edge shall be within 0.05 mm maximum tolerance zones. See Figure 6.

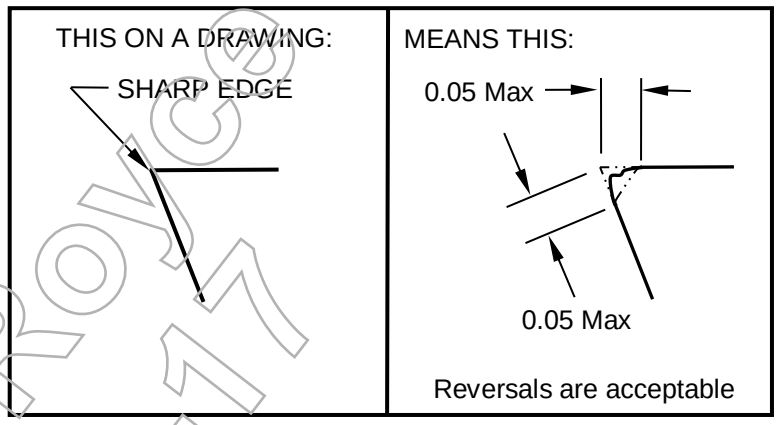

FIGURE 6 – SHARP EDGE

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10.3 Break Sharp Edge:

10.3.1 Where “Break Sharp Edge” is specified, the actual surface can take any form from a chamfer to a radius but must be within the edge break tolerance zone.

10.3.2 For surfaces or segments of surfaces that meet at nominal angles of 15º or less it is not necessary to break the edge. However, all loose or adhering burrs shall be removed. See Figure 7. This also applies to secondary edges such that an edge break could be made up of numerous segments, each meeting at angles of 15° or less.

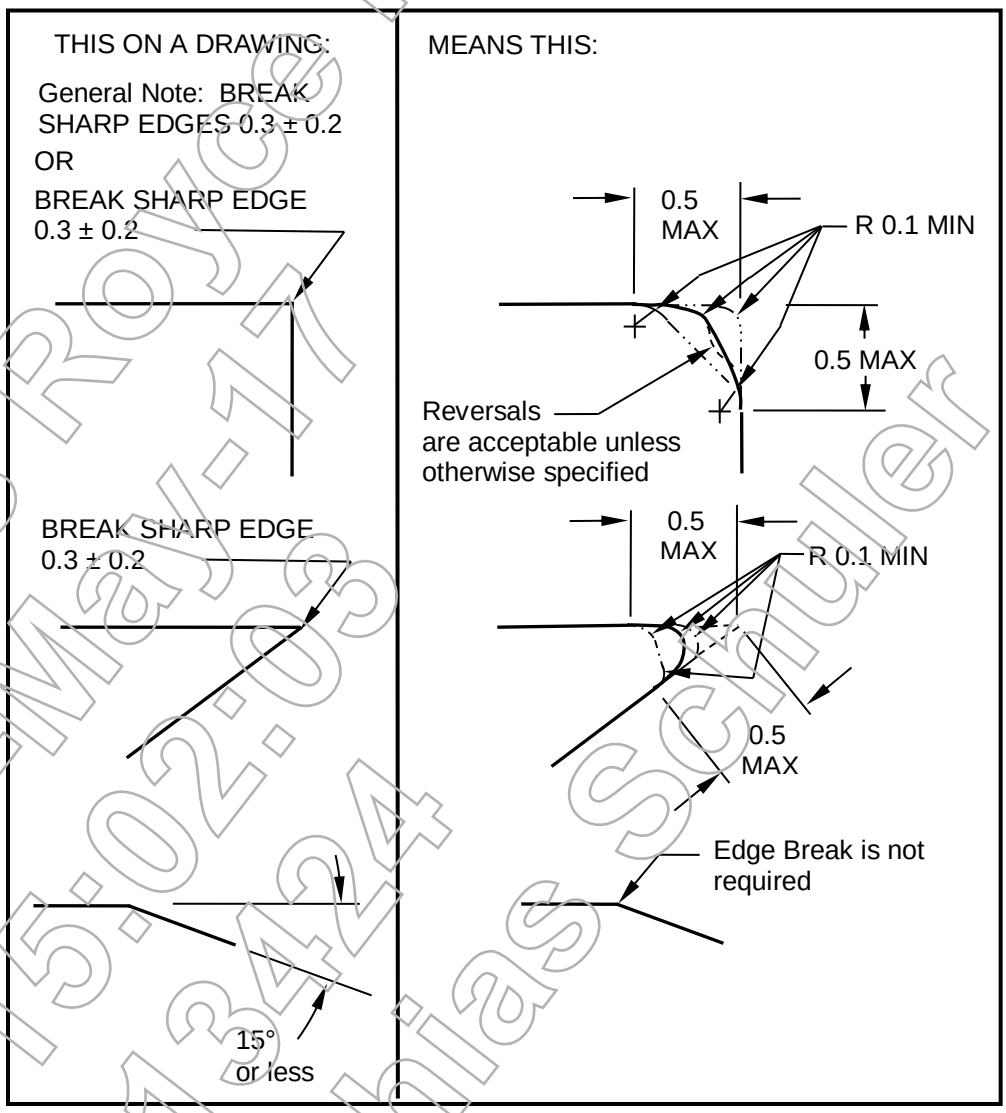

FIGURE 7 – EDGE BREAK

10.3.3 Edge breaks produced to comply with a default general note in the Component Definition title block as BREAK SHARP EDGES X.X ± X.X UOS are considered non-functional and are may be inspected visually.

10.3.4 Edge breaks produced to comply with a local note in the body of the definition are considered functionally important. Therefore, the edges identified by the local note require inspection operations.

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10.3.5 Methods used to break sharp edges shall not create new sharp edges.

10.3.6 For parts classified as Critical or Reliability Sensitive Tracked Life having a potential critical sensitivity to hand dressing shall have specified finished part features (e.g. break sharp edges) that are created by automated techniques only. Reference shall be made to the Fixed Process requirements for these parts.

10.3.7 For parts not classified as Critical or Reliability Sensitive Tracked Life, any method to break sharp edges may be used.

10.4 Radius:

10.4.1 Radius 'R'

10.4.1.1 A radius 'R' specifies the surface contour shall lie within a crescent shaped tolerance zone defined by the maximum and minimum radius including two lines defined by the adjacent surfaces See Figure 8.

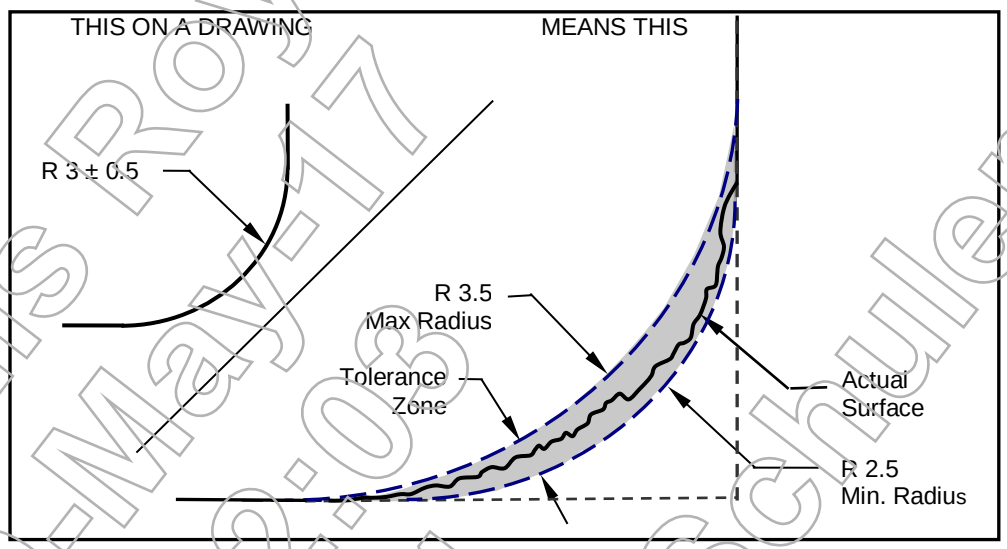

FIGURE 8 – RADIUS 'R'

10.4.2 Controlled Radius 'CR'

10.4.2.1 A controlled radius 'CR' specifies the surface contour shall lie within a crescent shaped tolerance zone defined by the maximum and minimum radius including two lines defined by the adjacent surfaces. Also, the surface curvature shall be visually tangent to the adjacent surfaces and visually smooth. See Figure 9. Examples of unacceptable surface conditions are shown in Figure 10.

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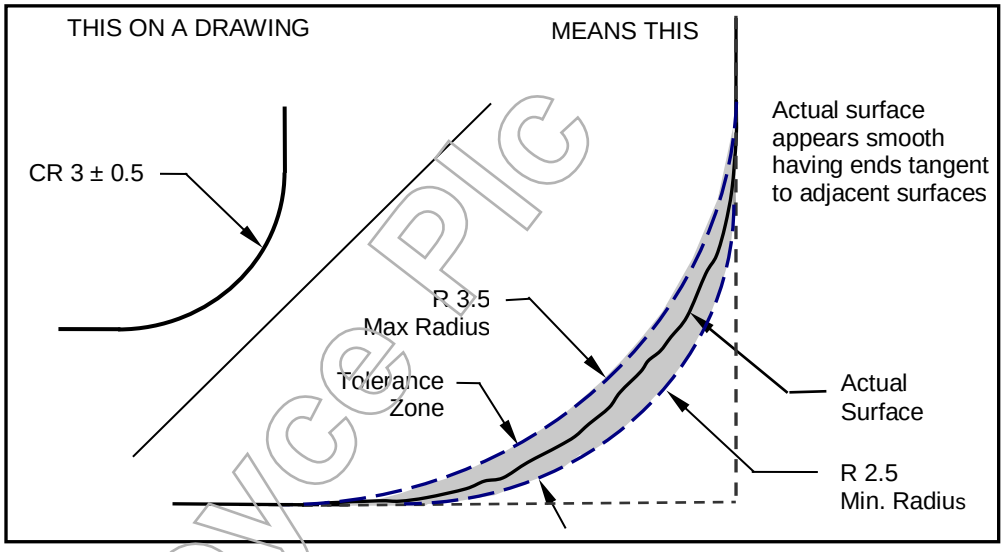

FIGURE 9 – CONTROLLED RADIUS 'CR'

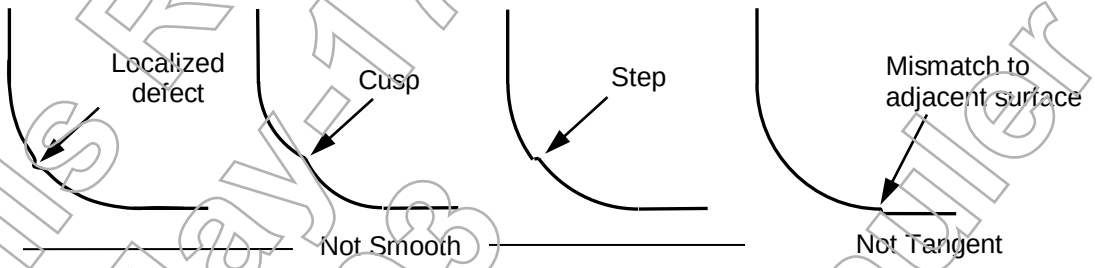

FIGURE 10 – EXAMPLES OF UNACCEPTABLE SURFACE CONDITIONS FOR 'CR' RADIUS

10.4.3 Basic Radius with Profile Control:

10.4.3.1 Per ASME Y14.5M-1994.

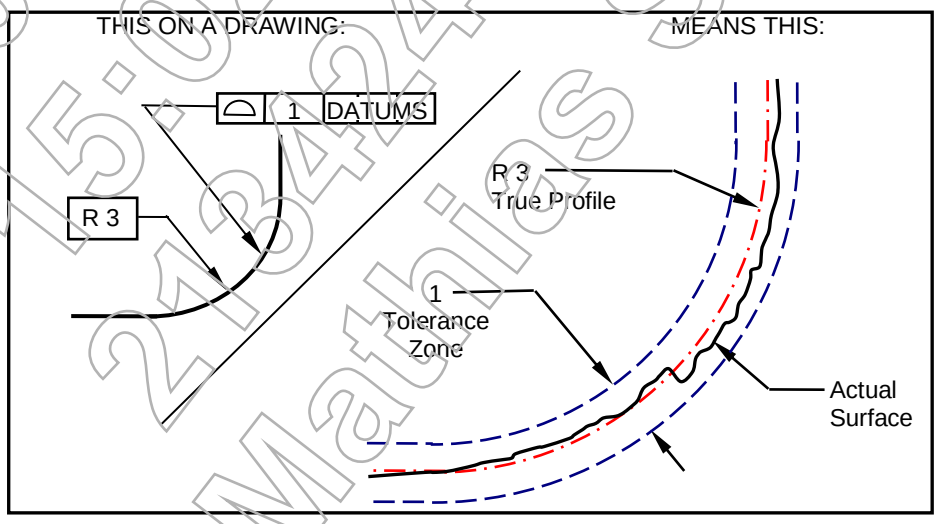

FIGURE 11 – EXAMPLE OF A BASIC RADIUS WITH SURFACE PROFILE CONTROL

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10.4.4 Basic Radius with Profile Control and Form Control

10.4.4.1 This describes a radius where the surface contour is controlled by “profile of a surface” and “profile of a line per unit length” geometric requirements. See Figure 12. “Profile of a surface” is interpreted per ASME Y14.5M-1994. “Profile of a line per unit length” is a refinement of the profile of a surface tolerance and is interpreted as follows:

All elements of the actual surface shall lie within a floating arc sector tolerance zone. The width of the zone (W) is equally disposed about an arc whose curvature is equal to the basic radius. The length of the zone (L) is an arc length measured along the mean radius. The actual surface can extend out the ends of the arc sector tolerance zone but cannot violate the inner or outer radius boundaries of the arc sector. See Figure 12. The arc sector tolerance zone is not located or oriented relative to the basic radius but floats freely along the actual surface. These requirements apply independently at each cross section of the radius.

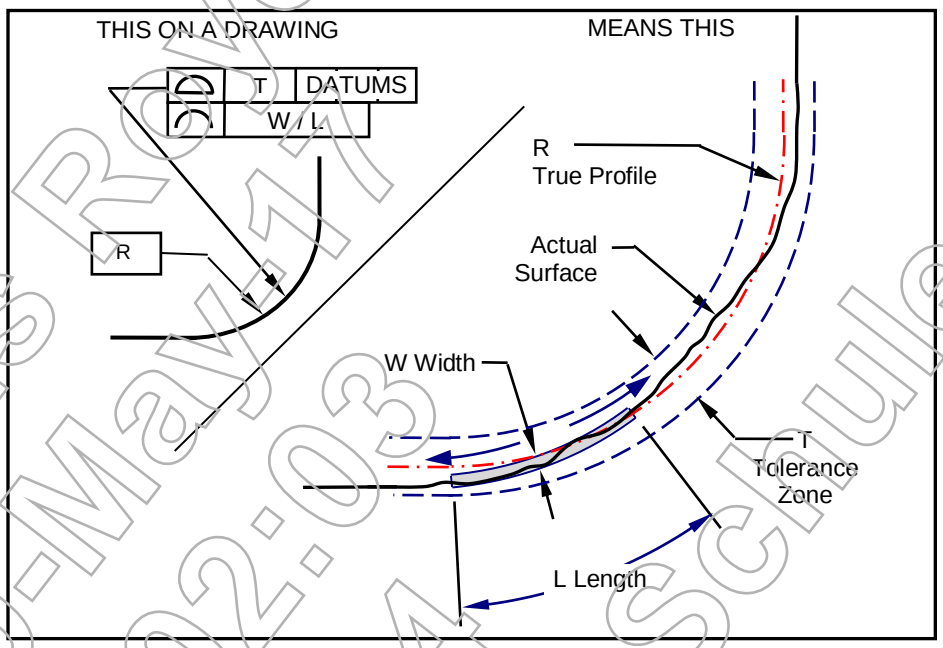

FIGURE 12 – BASIC RADIUS WITH PROFILE CONTROL AND FORM CONTROL

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10.5 Chamfer:

10.5.1 There are two methods used in specifying a chamfer: An angle and a linear dimension or two linear dimensions (See Figure 13). The actual surface of a chamfer shall be located within the tolerance zones.

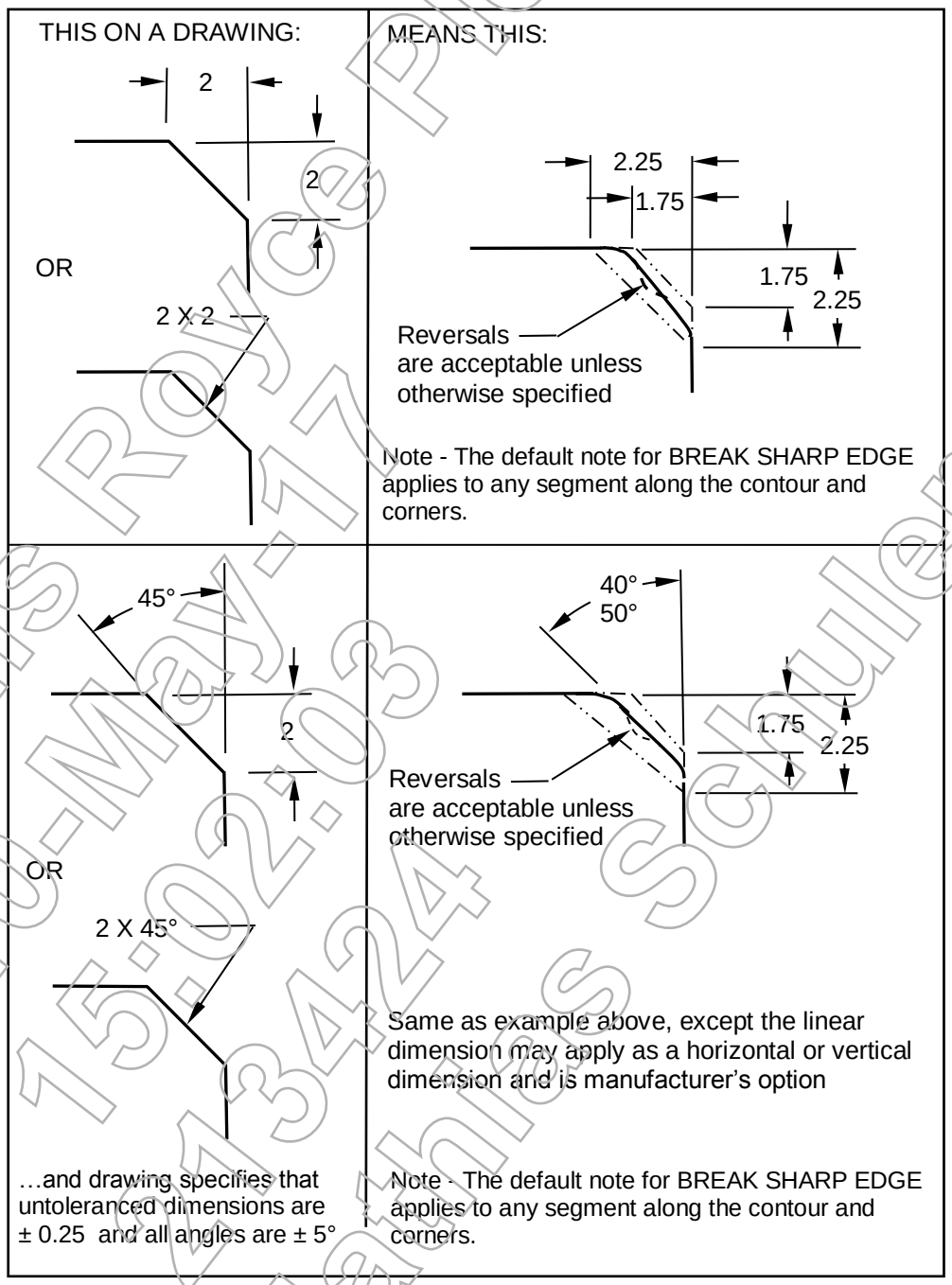

FIGURE 13 - CHAMFER

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10.6 Blend:

10.6.1 When a blend note is specified the actual surfaces within and between tolerance zones shall have visually smooth transitions. See Figure 14.

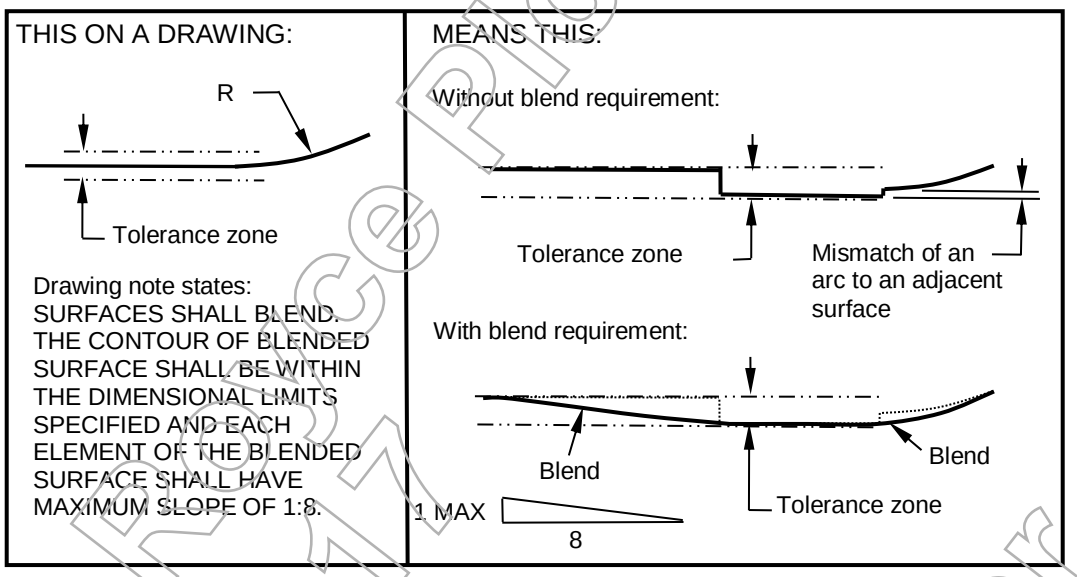

FIGURE 14 - EXAMPLE OF BLENDING OF SURFACES SPECIFIED AS A NOTE

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11 THREADED HOLES:

11.1 Where a depth is listed associated with a screw thread designation of a threaded hole, that depth defines the region where full threads are required. Beyond that depth, thread runout (i.e. imperfect or incomplete threads) is permissible. See Figure 15.

12 BLIND HOLES:

12.1 When a depth is specified for a circular hole, a drill point is permissible beyond that depth, unless otherwise specified. See Figure 15.

12.2 The drill point at the bottom of a blind circular hole shall have an included angle between 113° and 145°, unless otherwise specified. See Figure 15.

12.3 The inside corner radius for a flat bottom circular blind hole (i.e. no drill point) is 0.75 ± 0.25 mm, unless otherwise specified. See Figure 15.

12.4 The use of 'W' tip drills is not acceptable, unless otherwise specified.

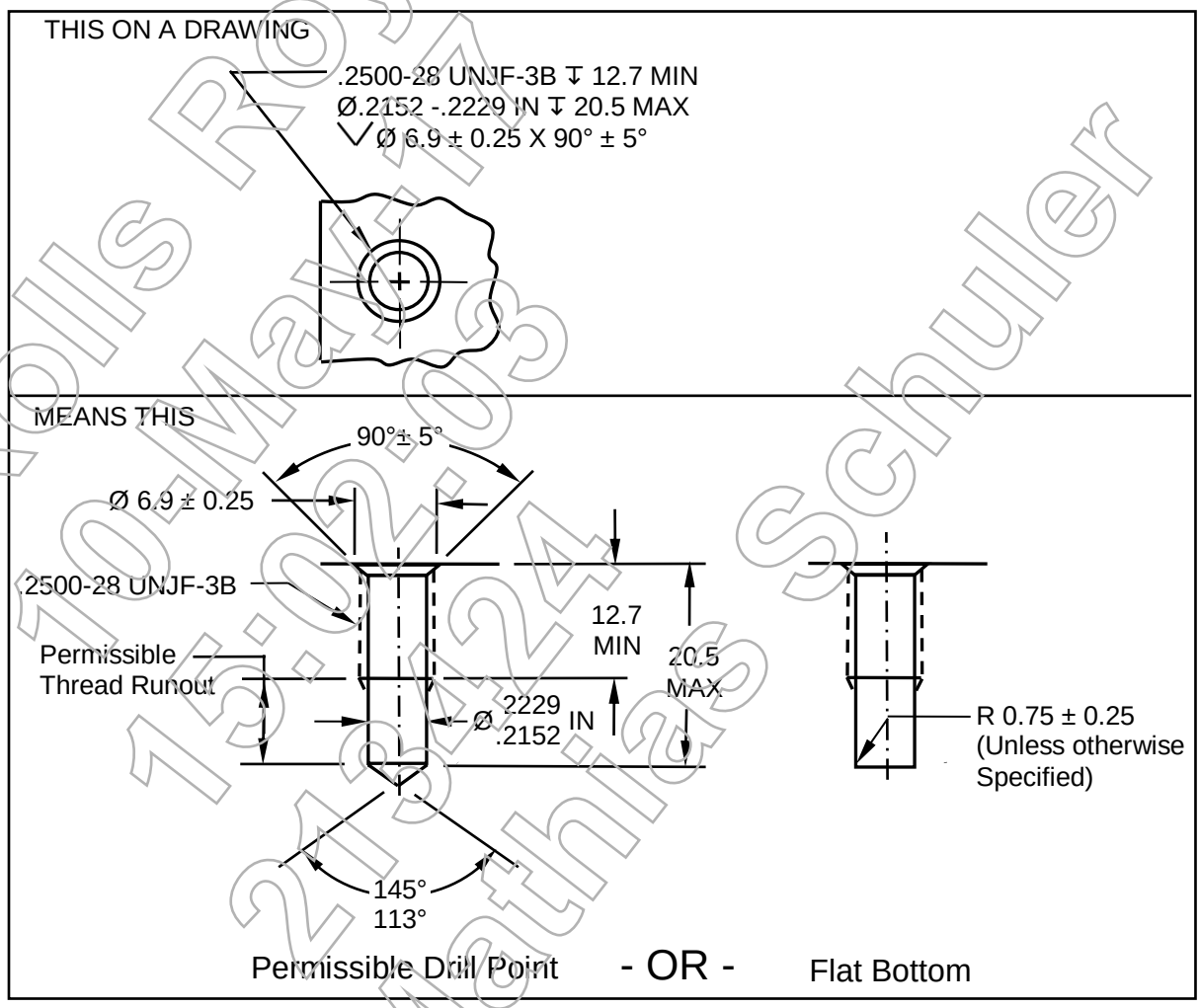

FIGURE 15 - EXAMPLE OF THREAD RUNOUT AND BLIND HOLE END CONFIGURATION

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13 NOTES:

13.1 Revision bars in the left margin indicate a change in this specification from the previous issue.

Issue	Change (ESC) Number	Change Summary (in descending order)	Date
C	ESC500028165335	Changed "Drawing" to "Definition" throughout the document. Para A3.2.1 should be Figure A2 Para 3.1 – For cancelled specifications, user shall notify Rolls-Royce. Para 3.2 and 3.3 remove "year issued": from specification numbers where they are not required. Para. 4.1 added definition of Definition. Para 4.2 added acronyms, KN, KSI, M, mm, MPa, NM Add new section 11 for KCF, KPC and CCF on Definitions Figure 15 change the order of the hole details for the top view of the blind threaded hole. Figure A2 clarify title.	17 Jan 2017
B	ESC500016365575	Clarify the requirements of radius callouts including redefining ASME Y14.5 "CR" and adding a basic radius defined with profile and form control. Clarify the configuration options for the end of blind hole. Change drill point angle limits. Clarify internal screw thread depths and permit thread runout. Add an Appendix for radius inspection guidance.	19 Jun 2012
A	ESC500012626091	This change adds more specific acceptance criteria for sharp edge, edge breaks, radius, chamfers, and introduces an interpretation of the term blend. Moved Section on angle of drill points from 11.3 to 12. Also clarified statement that "The ends of holes "machined with standard twist drill bits" shall have an included angle between 113° and 145° unless otherwise specified."	18 Mar 2010
Initial		This global specification is needed for the XWB engine program.	26 Mar 2009

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NONMANDATORY APPENDIX A – RADIUS INSPECTION

A1 This nonmandatory appendix gives guidance on common inspection methods which may be used to verify that a radius conforms to the requirements of Section 11.4.

A2 Four types of radius are defined in Section 11.4.:

Type 1 - Radius 'R'
Type 2 - Controlled Radius 'CR'
Type 3 - Basic Radius with Profile Control
Type 4 - Basic Radius with Profile Control and Form Control

Figure A1 shows these radius types and their associated inspection methods.

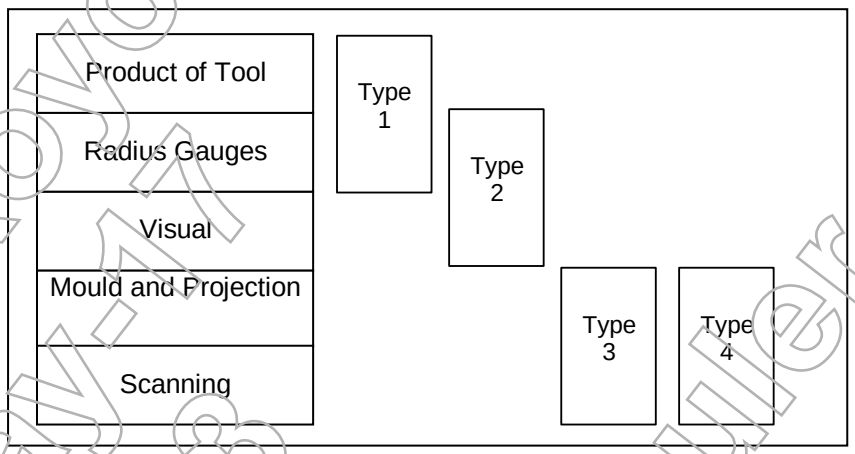

FIGURE A1 – RADIUS TYPES AND THEIR ASSOCIATED INSPECTION METHODS

A3 Inspection Methods:

A3.1 Product of Tool:

A3.1.1 Product of Tool is achieved by measuring the radii of the tooling before and after machining and by demonstrating the NC program is validated (usually first article inspection) and that the process is in control. It is essential that documentary evidence of these control measures is in place and a regular product audit programme demonstrates control measures are robust.

A3.2 Radius Gauges:

A3.2.1 Radius gauges (Figure A2) are used to evaluate the radius using visual comparison against the gauge. At a minimum, the radius should be checked with a gauge representing the upper and lower limits. For example, to check a radius of 5mm +/- 1mm, gauges of 4mm and 6mm should be used. If it is not possible to obtain radii gauges at the limits of tolerance, then the nearest gauge within the tolerance band should be used. For example for a radius of 5mm +/- 0.8mm gauges of 5.75 and 4.25 may be used. Radii measured outside of this band may still be conforming but would need to be verified with another method.

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A3.2.2 Gauge R&R or equivalent studies are typically a requirement for verification of the capability of a measurement system. Where radii gauges are used to measure radii, it is not usually practical or possible to verify their capability with such studies. In this case, the exception should be documented and approved with the local technical authority and quality function.

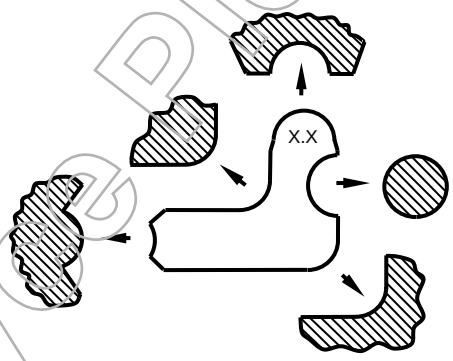

FIGURE A2 - TYPICAL RADIUS GAUGE AND ITS USE

A3.3 Visual:

A3.3.1 Visual is an exam nation of the radius surface to verify its form. This examination is typically accomplished without magnification. Radius gauges can assist with this process by providing a standard form for comparison. Visual inspection, in this context, is not to be used to verify the size of a radius.

A3.4 Mould and Projection:

A3.4.1 Mould and Projection involves optical projection to assess size and form. Radii can be directly projected or, where access prevents this, replica casts or moulds can be taken using suitable validated commercially available compounds. The moulds can be projected directly or be sectioned and projected.

A3.4.2 The projection method, moulding method, and compound should be validated against an independent technique and approved in accordance with local measurement system validation processes. Validation of the technique should as a minimum consider the following:

- Dimensional stability of the shape of the mould
- Material compatibility of the compound
- Projection method
- Sectioning technique (if applicable) and any errors introduced

A3.5 Scanning:

A3.5.1 Scanning involves using digital verification methods including, but not limited to, coordinate measurement machines (CMM), laser scanners, or contour tracers. These can either scan the radius directly or scan a mould of the radius.

A3.5.2 Scanning methods should be validated against an alternative independent technique to demonstrate capability and approved in accordance with local measurement system validation processes. Particular attention should be paid to demonstrate the scanning method can detect below-minimum radii sizes. A common failure mode of these systems is to represent very small radii as larger than they actually are.

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